The High-Redshift Accretion Atlas

A reproducible JWST catalogue and growth-model comparison at $z \geq 4$

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Abstract

We present a provenance-aware catalogue and visual growth atlas for JWST-identified accreting black holes and candidates at $z \geq 4$. The final v3 catalogue contains 350 literature measurements representing 340 physical objects in 339 host systems and 32 source families. Of these, 237 objects have a canonical, method-comparable numerical black-hole mass and enter the growth analysis; the remaining 103 are retained as explicit no-inference records. We compare each eligible object with an analytic exponential-growth model, propagate published asymmetric statistical mass errors with 10,000 deterministic Monte Carlo draws, and preserve mass-method systematics as separate envelopes. Under the reference assumptions $z_{\rm seed} = 30$, radiative efficiency $\epsilon = 0.1$, no merger boost, and a $10^2\,M_\odot$ seed, 14 objects require a lifetime-average $f_{\rm Edd} > 1$ at their point estimates. Ten remain above unity at the 16th percentile, and eight have $P(f_{\rm Edd} > 1) \geq 0.95$. UNCOVER-20466 is the highest-pressure object, with median required $f_{\rm Edd} = 1.453$ and a 16th–84th percentile interval of 1.354–1.549. Because the input samples have different selection functions and mass estimators, all global rankings are navigation aids rather than demographic measurements. The released catalogue, source provenance, analysis tables, figures, and 680 per-object panels reproduce through an ordered five-notebook workflow.

1 Introduction

JWST spectroscopy has revealed broad-line, narrow-line, high-ionization, X-ray, and photometrically selected accreting black-hole candidates within the first billion years. Their inferred masses and early observation times constrain how rapidly black-hole seeds must grow, but cross-paper comparison is difficult. Published samples use different selection functions, evidence thresholds, virial calibrations, lensing treatments, and definitions of compact or red sources. The same physical object may also appear under different identifiers or with multiple measurements.

The High-Redshift Accretion Atlas addresses this comparison problem at the level of individual objects. It standardizes source-native measurements, resolves identities, records evidence and mass provenance, and evaluates every supported mass against the same family of analytic growth histories. The atlas asks two bounded questions: which objects place the strongest pressure on a stated growth model, and which measurements would most change that assessment? It does not estimate a pooled mass function, occurrence rate, or formation-channel fraction.

The input literature spans JWST broad-line samples [2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 34, 35], host-selected broad-line candidates [9], an X-ray evidence history [18, 20], JADES narrow/high-ionization candidates [21], and GN-z11 high-ionization evidence [22]. Heterogeneous additions include high-ionization and narrow-line candidates [23, 24, 25, 29, 30], MIRI SED-selected candidates [26, 27], GHZ9 [28], compact blue broad-line emitters [31], additional UNCOVER candidates [32], MoM-BH*-1 [33], and GHZ4/GHZ7 [36].

2 Catalogue construction

2.1 Scope and versioning

The catalogue uses scientific dataset versions rather than software releases. v1 is the original 23-object JADES broad-line catalogue. v2 expands the same mass-comparable analysis to 211 JWST broad-line objects. v3 retains all v2 objects and adds heterogeneous JWST-identified evidence classes. A source enters only one admission version; an internally heterogeneous source is assigned in full to v3. The fixed scope requires $z \geq 4$ and identification or candidate classification from JWST data. Objects merely followed up by JWST after a non-JWST discovery are outside the admission rule.

The v3 catalogue has 350 measurement rows, 340 physical objects, and 339 host systems. It contains 247 broad-line AGN, 50 narrow-line candidates, 31 photometric candidates, ten high-ionization candidates, and two X-ray candidates. Preferred object-level evidence comprises 214 secure, 41 probable, 84 candidate, and one disputed classification. These labels describe the source papers' evidence and do not place all classes on a single purity scale.

2.2 Provenance, extraction, and identity

Each measurement carries stable measurement, physical-object, and host-system identifiers. Source-native values, adopted standardized values, uncertainty semantics, table locations, URLs, DOI or archive versions, archive SHA-256 hashes where available, extraction dates, and caveat tags are retained. The manual-extraction audit covers 26 immutable row-level artifacts, while the provenance registry contains 36 records covering all 32 admitted source families. Auxiliary coordinate and contextual sources are attributed separately to the THRILS catalogue [8], the UHZ1 spectroscopy paper [19], and the versioned JADES DR3 GOODS-S prism catalogue.

Identity resolution preserves every admitted measurement while selecting one preferred row for object-level inference. Alternate measurements are never silently averaged. The final completion illustrates this policy: 15 NEXUS NIRCam/WFSS measurements [34], 13 COSMOS-3D NIRCam-grism measurements [35], and GHZ4/GHZ7 [36] add 30 measurements but 29 physical objects. NEXUS source NX10835 lies 0.066 arcsec from an existing Mascia identifier at consistent redshift, so the new mass-bearing measurement becomes preferred for the same object.

The Scholtz et al. source audit provides another cardinality check. The paper reports 42 objects, whereas its source-native table contains 41 rows. Exactly 21 table rows satisfy $z \geq 4$; all are admitted. One is identity-linked to a broad-line object, yielding 20 distinct narrow-line candidates from that source. JADES-NS-GS00099671 is retained with its published host and bolometric observables but without an invented black-hole mass.

2.3 Mass comparability and analysis subsets

A measurement enters numerical growth analysis only when the source publishes a canonical numeric black-hole mass compatible with the declared comparison policy. Conditional masses inferred by assuming an Eddington ratio, indirect model ranges, and upper limits remain contextual observables. This produces 244 eligible measurements and 237 eligible preferred objects. Of those objects, 236 use Balmer-line single-epoch virial masses and one uses a UV single-epoch virial mass. Their redshifts span 4.000–10.603 with median 5.110; standardized $\log_{10}(M_{\rm BH}/M_{\odot})$ spans 5.51–9.31 with median 7.18. The 103 objects without a comparable mass remain visible throughout the catalogue and atlas.

Table 1: v3 scale and analysis subsets.

Product or subset	Rows or objects
Measurements / physical objects / host systems	350 / 340 / 339
Source-local observables	810
Growth-eligible measurements / objects	244 / 237
Primary measurements / objects	234 / 227
Catalogue-only physical objects	103
Alternate-measurement sensitivity pairs	7
Source-family caveat rows	32
Per-object parameter-map panels	680

3 Growth-model comparison

3.1 Reference model

We use the analytic exponential-growth relation adopted by Dayal [1],

$$M_{\rm BH}(t_{\rm obs}) = M_{\rm seed} \exp\left[f_{\rm Edd} \left(\frac{1-\epsilon}{\epsilon} \right) \frac{t(z_{\rm obs}) - t(z_{\rm seed})}{t_{\rm Edd}} \right] B_{\rm merge}, \tag{1}$$

where $t_{\rm Edd} \simeq 0.45$ Gyr, ϵ is a constant radiative efficiency, and $B_{\rm merge}$ is an effective multiplicative merger boost. For every eligible object we invert Eq. 1 to obtain the required lifetime-average $f_{\rm Edd}$ at fixed seed mass and the required seed mass at fixed $f_{\rm Edd}$. The reference ranking uses $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$, and $M_{\rm seed} = 10^2\, M_\odot$.

These parameters summarize a growth history rather than reconstruct it. In particular, the lifetime-average $f_{\rm Edd}$ is not the object's currently observed Eddington ratio. Constant efficiency, uninterrupted exponential growth, and a single merger factor omit feedback cycles, changing accretion states, and stochastic mergers. We therefore call the derived quantity *growth pressure*: it measures how demanding a specified history is, conditional on the published mass and the reference model.

The spin-dependent $f_{\rm Edd}$–seed-mass maps and compatibility matrices use a different, illustrative efficiency prescription. For $f_{\rm Edd} \leq 1$ they use the ideal Kerr thin-disk efficiency $\epsilon_a = 1 - E_{\rm ISCO}(a)$, approximately 0.038, 0.057, and 0.423 for $a = -1, 0, +1$. Above unity they use

$$\epsilon_{\rm eff} = \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}). \tag{2}$$

This follows from the phenomenological photon-trapping relation $L/L_{\rm Edd} = 1 + \ln \dot{m}$, with

$$\dot{m} = \frac{\dot{M}}{L_{\rm Edd}/(\epsilon_a c^2)}. \tag{3}$$

It is a parameter-scan prescription, not a relativistic slim-disk or spin-evolution calculation. The baseline rankings, duty-cycle diagnostics, and seed-redshift–mass maps retain $\epsilon = 0.1$; growth tracks retain their stated constant efficiencies. Thus compatibility above unity is conditional on a different efficiency model from the baseline required-$f_{\rm Edd}$ ranking.

3.2 Uncertainty and ranking policy

Published asymmetric statistical uncertainties in $\log_{10} M_{\rm BH}$ are propagated with 10,000 deterministic split-side draws per measurement and per preferred object. If a source gives no statistical error, the atlas does not invent one. In particular, the twelve mass-bearing NEXUS objects have no published statistical mass errors. Their repeated point values yield zero-width stored intervals; probabilities and uncertainty ranks are left unavailable, not interpreted as certainty. They are marked separately in uncertainty figures. The other 225 objects have reported statistical errors. Split-side sampling assigns half the probability to each side of the reported central value, using the respective error as the scale of a half-normal distribution; it approximates quoted intervals rather than reconstructing source posteriors. Reported virial-calibration or method-scale systematics are stored and, where possible, mapped to separate sensitivity envelopes; they are not combined with statistical errors as though they were independent Gaussian terms.

The primary scientific comparison is within object class and mass-comparability group. Global point and uncertainty ranks are retained only for navigation. The point navigation score is

$$S = \min\left[100, 100\max\left(C\left(\frac{f_2 - 0.3}{1.2}\right), C\left(\frac{s_{0.3} - 4}{2.8}\right)\right) + 8C\left(\frac{z-6}{4}\right)\right], \tag{4}$$

where $C(x) = \min(1, \max(0, x))$, f_2 is the required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed, and $s_{0.3}$ is the required $\log_{10}(M_{\rm seed}/M_\odot)$ at $f_{\rm Edd} = 0.3$. This heuristic combines two growth diagnostics and a redshift term; it is not a probability. Ties use decreasing f_2, decreasing redshift, then stable identifier. The manuscript's required-$f_{\rm Edd}$ table is instead ordered directly by f_2. Compatibility is also calculated object by object across four seed families, three spin/efficiency cases, two merger cases, and three average accretion rates. The source-level selection registry leaves inverse-probability weights unset because no common inclusion-probability model spans the 32 source families. Consequently, compatibility fractions are descriptive summaries of the admitted objects, not population frequencies.

4 Results

4.1 Growth pressure under the reference assumptions

The required $f_{\rm Edd}$ distribution for a $10^2\,M_\odot$ seed has median 0.574 and interquartile range 0.488–0.680. Fourteen of 237 eligible objects require $f_{\rm Edd} > 1$ at their point estimates; none requires $f_{\rm Edd} > 2$. The high-pressure tail is not simply a mass ranking because less cosmic time can offset a lower observed mass. GN-z11, for example, has a lower standardized mass than many broad-line objects but ranks second because it is observed at $z = 10.603$.

UNCOVER-20466 leads both the point and uncertainty navigation views. Its point-estimate required $f_{\rm Edd}$ is 1.452, and its Monte Carlo median is 1.453 with a 16th–84th percentile interval of 1.354–1.549. The newly admitted COSMOS3D-13852 ranks third by the composite navigation score and fourth by required $f_{\rm Edd}$. GS-20057765 has the third-highest required $f_{\rm Edd}$. Table 2 reports the five largest point-estimate required $f_{\rm Edd}$ values, independent of the composite score.

Table 2: Highest point-estimate required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed under the reference assumptions. Intervals propagate published statistical mass errors only.

Object	z	$\log_{10} M_{\rm BH}$	Required $f_{\rm Edd}$	16th–84th percentile
UNCOVER-20466	8.500	8.17	1.452	1.354–1.549
GN-z11	10.603	6.20	1.437	1.335–1.542
GS-20057765	8.913	7.33	1.355	1.177–1.513
COSMOS3D-13852	7.646	8.60	1.314	1.247–1.384
RUBIES-EGS-55604	6.986	9.31	1.267	1.250–1.284

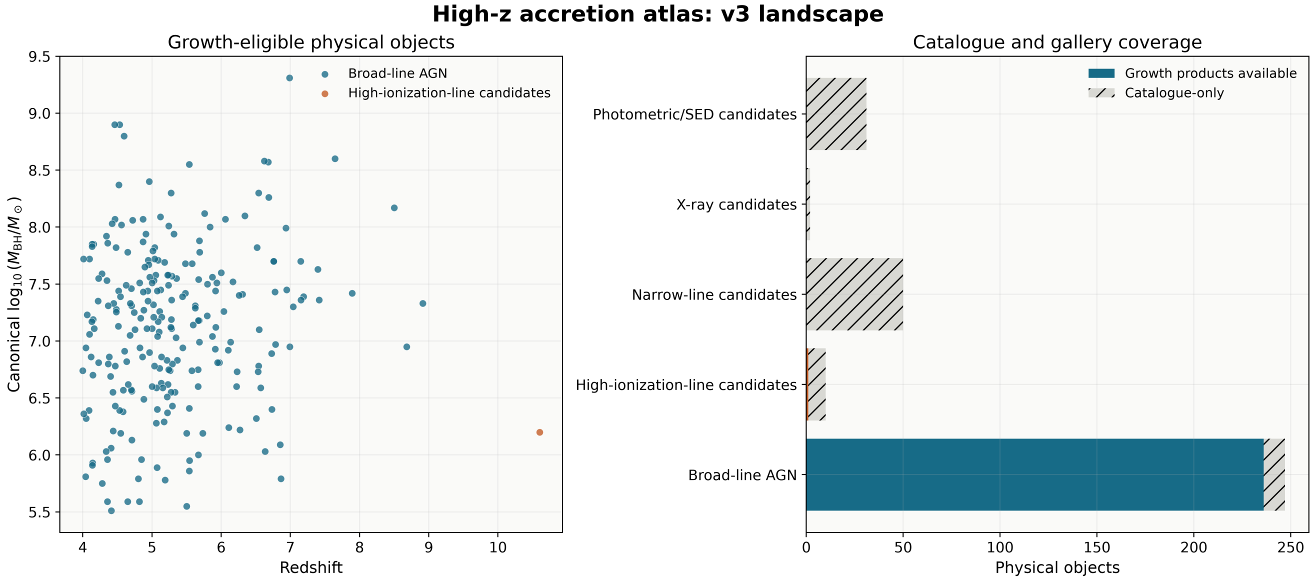

Figure 1: Catalogue landscape. Left: the 237 growth-eligible physical objects in black-hole-mass–redshift space. Right: product coverage for all 340 objects. The 103 catalogue-only objects remain available for provenance and evidence analysis without unsupported numerical growth products.

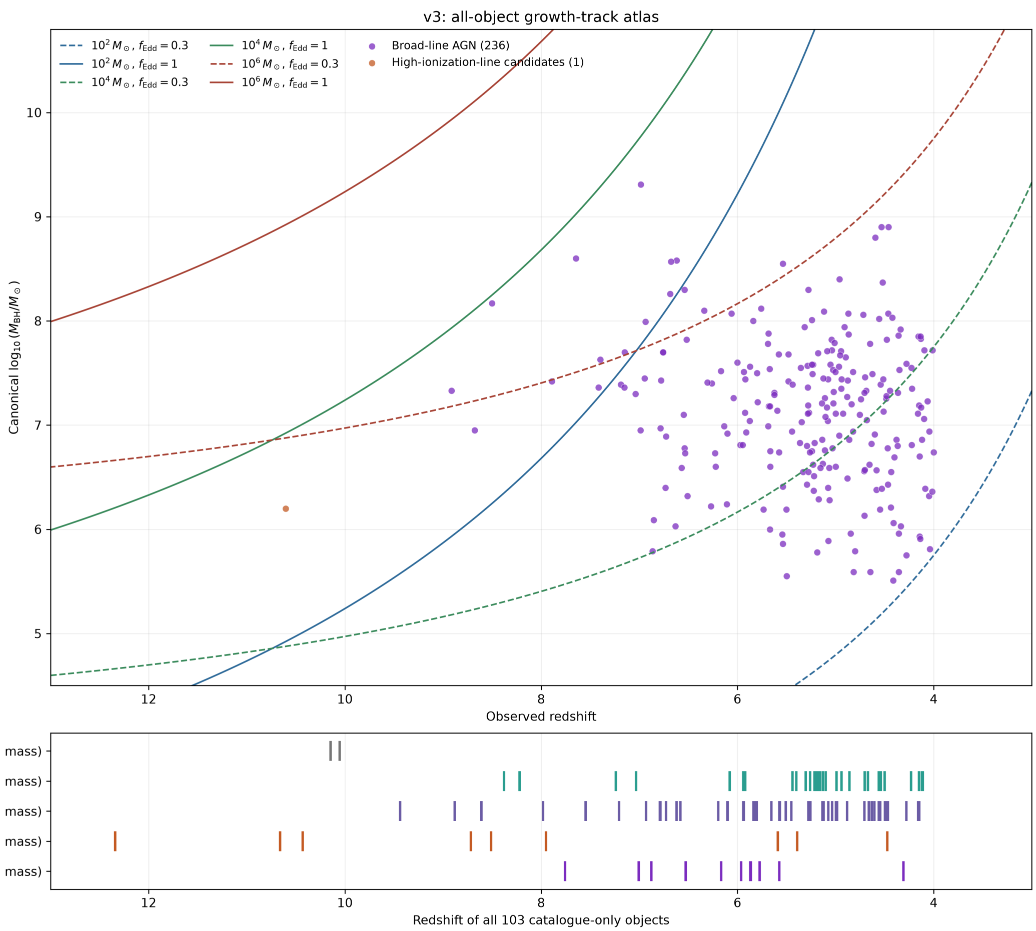

Figure 2: All-object growth-track view. The upper panel compares the 237 supported masses with reference tracks; the lower panel retains the redshift and class of all 103 no-mass objects. The reversed redshift axis spans $z = 13$ to $z = 3$.

4.2 Robustness to reported statistical uncertainty

The Monte Carlo results preserve the point-estimate conclusion for the strongest objects. Fourteen objects have median required $f_{\mathrm{Edd}} > 1$, ten remain above unity at the 16th percentile, and 15 cross unity by the 84th percentile. Eight objects have $P(f_{\mathrm{Edd}} > 1) \geq 0.95$. These probabilities are conditional on the reported statistical errors and the reference growth assumptions; they do not include a shared virial-calibration uncertainty or uncertainty in seed redshift, radiative efficiency, and merger history.

4.3 Model compatibility and measurement choice

Compatibility changes substantially across the seed, efficiency, merger, and average-accretion grid. This dependence is the scientific point of the compatibility layer: an object's apparent tension cannot be assigned to seed mass alone without specifying the other assumptions. The complete matrix keeps unavailable cases distinct from incompatibility, preventing the 103 massless objects from being counted as failed growth histories.

Seven eligible objects have retained alternate measurements. Their sensitivity table reports how preferred-measurement choice changes standardized mass and required f_{Edd}. The follow-up matrix includes all 340 objects: ten are classified as robust high-pressure targets, seven as caveated high-pressure targets, 31 as uncertainty-leverage targets, 21 as comparison anchors, one as a source-consistency target, 167 as contextual records, and 103 as requiring a method-comparable mass before growth inference.

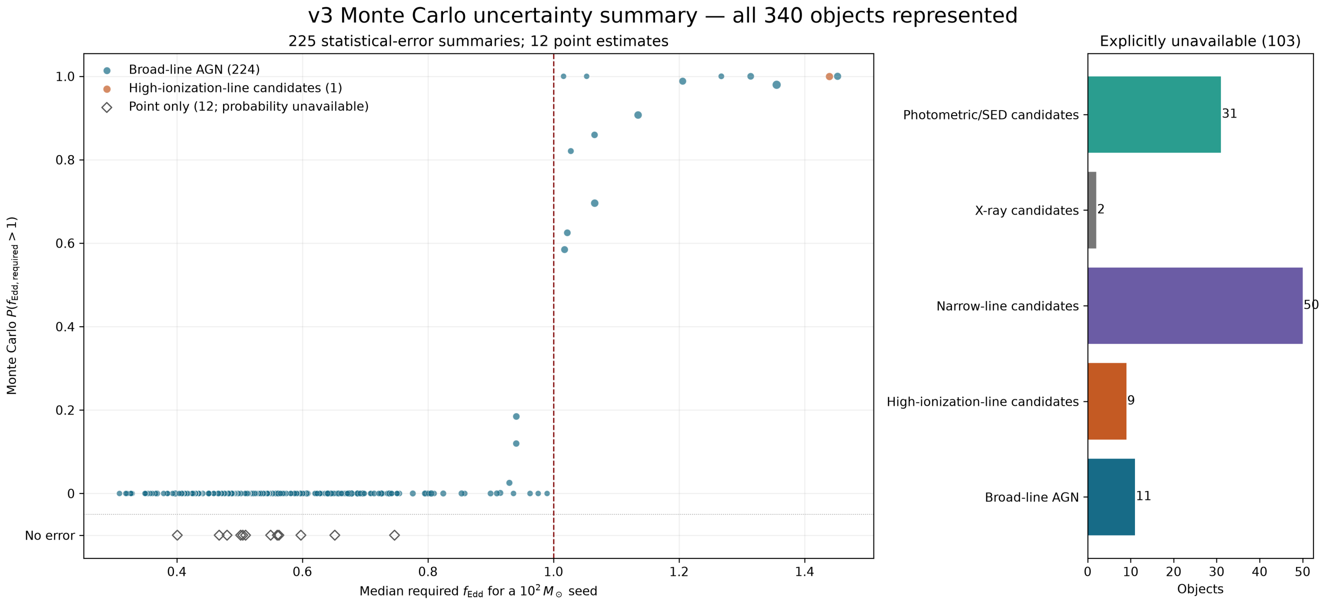

Figure 3: Monte Carlo uncertainty summary for all 237 numerical objects. Point size encodes the 16th–84th percentile width. The accounting panel retains all 103 catalogue-only objects.

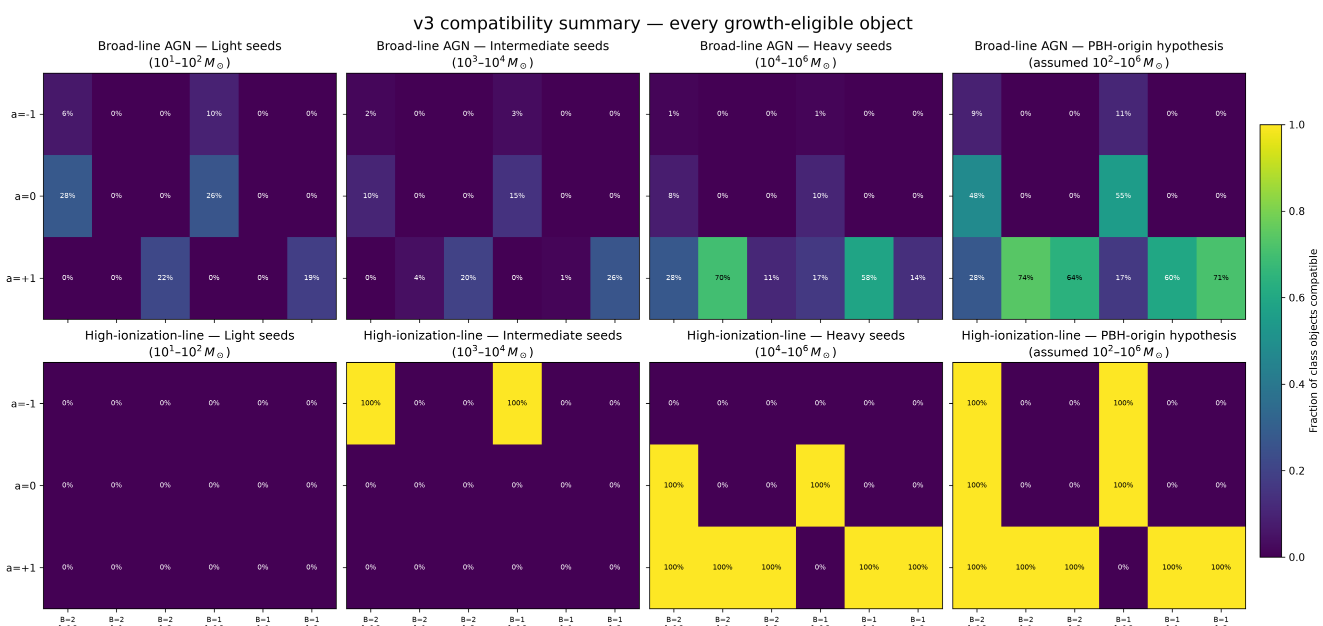

Figure 4: Descriptive compatibility across seed families, spin/efficiency cases, merger boosts, and average accretion rates. Percentages are within admitted classes and are not completeness-corrected demographic estimates.

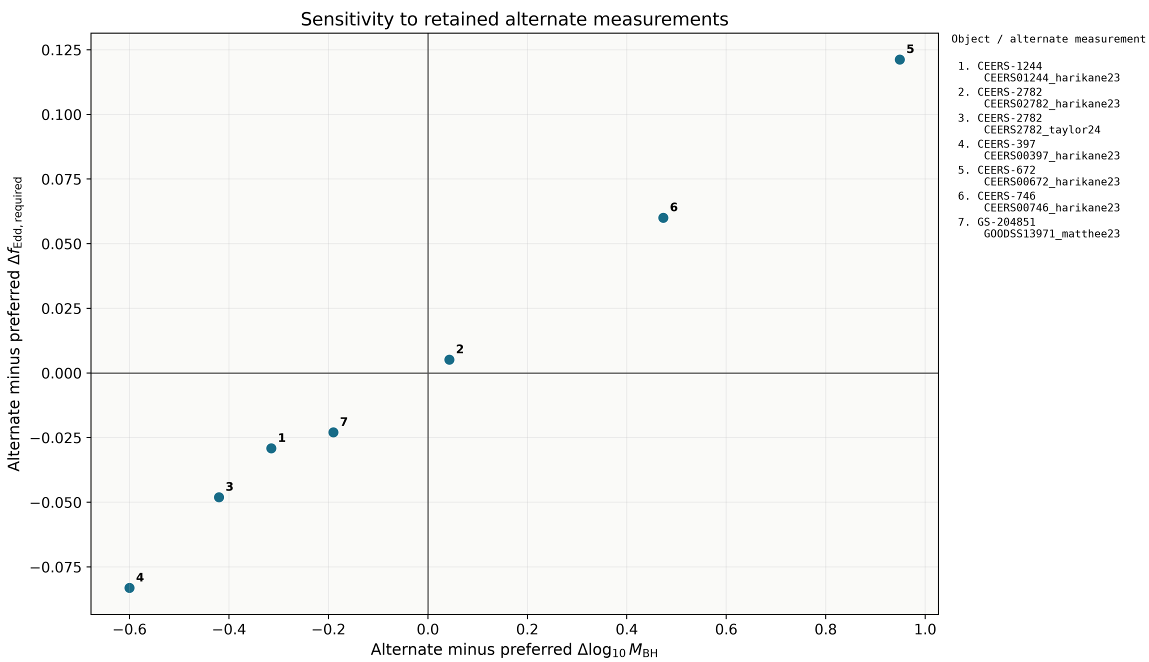

Figure 5: Sensitivity to seven retained alternate measurements. Numbered points are keyed to stable object and measurement identifiers.

5 Discussion

5.1 What the high-pressure tail establishes

The reference calculation identifies a small set of objects that are difficult to grow from light seeds under continuous radiatively efficient accretion. The result does not select a unique remedy. Earlier or heavier seeds, lower effective radiative efficiency, merger growth, or episodes above the reference accretion rate can each reduce the tension. The atlas is designed to expose these trade-offs for each object rather than collapse them into a single formation label.

The most useful targets are therefore those whose placement is both demanding and stable to published statistical uncertainty. Independent mass measurements, better separation of broad-line-region kinematics from scattering or outflows, and constraints on obscuration and lensing can change the physical interpretation more than another evaluation of the same analytic formula. The follow-up matrix makes that distinction explicit.

5.2 Value of the heterogeneous evidence layer

The 103 no-mass objects remain within project scope because the catalogue is a cross-paper evidence atlas as well as a growth calculator. Narrow-line, high-ionization, X-ray, and SED-selected candidates trace discovery channels that a mass-only broad-line catalogue would omit. They can be compared in redshift, evidence, survey, and follow-up need, but they are excluded from numerical growth claims until a suitable mass exists. This separation lets the atlas retain potentially important early accretors without converting qualitative evidence or model-dependent context into artificial precision.

5.3 Limitations

The dominant limitations are scientific rather than computational. The catalogue is a union of heterogeneous selections and cannot support pooled demographics without source-specific completeness models. Single-epoch virial masses share calibration and geometry uncertainties that are incompletely reported across papers. Some identities in crowded or lensed fields may ultimately benefit from probabilistic cross-matching. Finally, Eq. 1 compresses time-dependent accretion, feedback, mergers, and radiative-state changes into effective parameters. Its outputs should be read as conditional comparisons, not reconstructed histories.

6 Reproducibility and data products

The repository pins the direct Python 3.12 dependencies and provides five ordered notebooks that materialize catalogues, generate science tables, render figures, build the complete atlas, and verify the release. Canonical manifests record SHA-256 hashes for 85 v1, 461 v2, and 721 v3 artifacts. Verification checks strict nested version membership, catalogue cardinalities, exact in-memory CSV reproduction, 10,000-draw uncertainty products, compatibility coverage, figure resolution, and both parameter-map panels for every object. The current result inventory contains 1,237 artifacts.

Independent source-cell checks additionally compare 865 values across 53 admitted rows from the JADES, NEXUS, COSMOS-3D, and Seven Wonders tables. They recovered previously omitted NEXUS line-luminosity errors, now preserved in the source observables without changing masses or ranks. Numerical quadrature of the Friedmann age integral and Runge–Kutta growth integration using physical constants provide checks independent of the production closed-form functions. This supplements reproducibility; it is not an exhaustive independent re-audit of every measurement and interpretation across all 32 families.

Every v3 object has an $f_{\rm Edd}$–mass panel and a seed-redshift–mass panel. The 237 eligible objects receive numerical products; the 103 others receive explicit status panels. Combined all-object compatibility and Monte Carlo atlases retain every label for close inspection. A supplementary full-assumption growth-track figure preserves the historical v1 grid of three seed masses, three $f_{\rm Edd}$ values, four constant efficiencies, and two merger boosts, for 72 curves.

7 Conclusions

We have constructed a reproducible atlas of 340 JWST-identified accreting black-hole objects and candidates at $z \geq 4$. The main results are:

- 237 objects have a canonical method-comparable mass and 103 remain explicit no-inference records;
- under the light-seed reference history, 14 objects require point-estimate $f_{\rm Edd} > 1$, ten remain above unity at the 16th percentile, and eight have $P(f_{\rm Edd} > 1) \geq 0.95$;
- UNCOVER-20466 places the strongest pressure on that history, while the new COSMOS3D-13852 entry ranks third by the composite navigation score, while GS-20057765 is third by required $f_{\rm Edd}$;
- model compatibility is conditional on seed, efficiency, merger, and accretion assumptions, and the heterogeneous catalogue cannot support pooled demographic inference; and
- complete provenance, identity history, uncertainty semantics, source caveats, follow-up priorities, and per-object visual products remain linked to the same frozen v3 dataset.

The next scientific step is a selection-aware inference layer with explicit survey completeness, method-calibrated hierarchical uncertainty, and targeted follow-up tied to which observation would most change each object's growth interpretation.

References

[1] Dayal, P. 2024, *A&A*, 690, A182, doi:10.1051/0004-6361/202451481.

[2] Juodžbalis, I. et al. 2026, *MNRAS*, 546, stag086, doi:10.1093/mnras/stag086.

[3] Taylor, A. J. et al. 2025, *ApJ*, 986, 165, doi:10.3847/1538-4357/add15b.

[4] Matthee, J. et al. 2024, *ApJ*, 963, 129, doi:10.3847/1538-4357/ad2345.

[5] Lin, X. et al. 2024, *ApJ*, 974, 147, doi:10.3847/1538-4357/ad6565.

[6] Harikane, Y. et al. 2023, *ApJ*, 959, 39, doi:10.3847/1538-4357/ad029e.

[7] Davis, K. et al. 2026, arXiv:2602.23310v1.

[8] Hutchison, T. A. et al. 2025, THRILS redshift catalogue, arXiv:2512.12509v1.

[9] Ren, J. et al. 2025, *MNRAS*, 544, 211–233, doi:10.1093/mnras/staf1709.

[10] Greene, J. E. et al. 2024, *ApJ*, 964, 39, doi:10.3847/1538-4357/ad1e5f.

[11] Kocevski, D. D. et al. 2025, *ApJ*, 986, 126, doi:10.3847/1538-4357/adbc7d.

[12] Skyfire Collaboration 2026, arXiv:2609.00112v1.

[13] Larson, R. L. et al. 2023, *ApJL*, 953, L29, doi:10.3847/2041-8213/ace619.

[14] Killi, M. et al. 2024, *A&A*, 691, A52, doi:10.1051/0004-6361/202348857.

[15] Übler, H. et al. 2024, *MNRAS*, 531, 355, doi:10.1093/mnras/stae943.

[16] Baccus, C. et al. 2026, *ApJ*, 1006, 165, doi:10.3847/1538-4357/ae7de7.

[17] Fei, Q. et al. 2026, *ApJ*, 1003, 244, doi:10.3847/1538-4357/ae6248.

[18] Bogdán, Á. et al. 2024, *Nature Astronomy*, 8, 126, doi:10.1038/s41550-023-02111-9.

[19] Goulding, A. D. et al. 2023, *ApJL*, 955, L24, doi:10.3847/2041-8213/acf7c5.

[20] Zou, F. et al. 2026, UHZ1 full-data X-ray reanalysis, accepted by *ApJ*, arXiv:2603.24893v2.

[21] Scholtz, J. et al. 2025, *A&A*, 697, A175, doi:10.1051/0004-6361/202348804.

[22] Maiolino, R. et al. 2024, *Nature*, 627, 59–63, doi:10.1038/s41586-024-07052-5.

[23] Chisholm, J. et al. 2024, *MNRAS*, doi:10.1093/mnras/stae2199.

[24] Tang, M. et al. 2025, accepted by *ApJ*, arXiv:2505.06359v2.

[25] Mazzolari, G. et al. 2024, *A&A*, 691, A338, doi:10.1051/0004-6361/202451860.

[26] Leung, G. C. K. et al. 2026, submitted to *ApJ*, arXiv:2607.02666v1.

[27] Lyu, J. et al. 2024, *ApJ*, 966, 229, doi:10.3847/1538-4357/ad3643.

[28] Napolitano, L. et al. 2025, *ApJ*, 989, 75, doi:10.3847/1538-4357/ade706.

[29] Zhang, Z. et al. 2026, accepted by *ApJ*, arXiv:2506.04350v2.

[30] Chavez Ortiz, O. A. et al. 2025, revised 2026, arXiv:2511.03035v2.

[31] Mascia, S. et al. 2026, submitted to *A&A*, arXiv:2608.25021v1.

[32] Treiber, H. et al. 2025, *ApJ*, 984, 93, doi:10.3847/1538-4357/adc38f.

[33] Naidu, R. P. et al. 2026, *Nature*, 656, 329–333, doi:10.1038/s41586-026-10846-4.

[34] Zhuang, M.-Y. et al. 2025, NEXUS broad-line AGN sample, arXiv:2505.20393v1.

[35] Lin, X. et al. 2025, COSMOS-3D broad-line AGN sample, accepted by *ApJ*, doi:10.3847/1538-4357/ae1b9b.

[36] Napolitano, L. et al. 2025, *A&A*, 693, A50, doi:10.1051/0004-6361/202452090.